Week 6: Nuclear and Particle Physics SOLUTIONS

Discussion Questions

1 Key elements in discussion: size of atom (very rough) range 0.1 – 0.7 nm; nuclei size range: 2 fm – 15 fm; using $r = r_0 A^{\frac{1}{3}}$ (r_0 = 1.2 fm), refer to Coulomb forces (long range); strong force (short range); refer to p, n and e, angular momentum, quantized energy levels, *n, l, m_l, and m_s*. See text (Walker 11th Ed., §40.1) – some beyond scope of this course.

2 Range of the strong nuclear force (SNF) is $\sim 10^{-15}$ m. As SNF effects both charged *and* neutral particles, it is the dominant force at short range. However, the Coulomb force produces repulsion effects between same-charged particles, and operates over all distance ranges. So as more particles are added, and they become further from other like-charged particles, the effects of Coulomb repulsion increase, preventing larger constructions of single charged particles via the SNF. To hold the matter together, more neutrons help by feeling only the SNF and not the Coulomb force. Higher atomic number elements have more neutrons than protons to maintain stability, moving away from a linear relationship between *Z* and *A*.

3 *Binding energy* is the mass-energy difference between the mass-energy of the individual components of a nucleus and the mass-energy of the assembled nucleus; it is equal to the energy that would be required to 'unbind' the nucleus into individual protons and neutrons.

4 Fission is the splitting of high mass nuclei (e.g. U-235, Pu-239) to produce lighter nuclei with higher bonding energy; the difference in total binding energy before and after accounts for the energy release (*Q*) of the reaction. Similarly with fusion, lighter nuclei (e.g. two deuterons combine to form helium-4) result in greater binding energy after the fusion than before, and hence a positive value of *Q*.

5 Emitted products had a range of both energy and momentum; there was 'missing' energy and momentum; neutral because charges balanced - see text section 42.5; more detail in *Hyperphysics* and *Wikipedia*

6 Orbital mechanics as described by Kepler would expect the outer galactic arms to orbit slower than the inter regions of the galaxy, behaving similar to our solar system. As the rotation speeds do not decrease with distance, more (non-luminous) mass is likely present in the outskirts of each galaxy. More mass would likely be necessary to maintain the required force to hold such fast-moving bodies in orbit around the galaxy. The strongest candidate for this mass is (cold) Dark Matter, though some suggest other modifications to existing theory (no such modified theories adequately explain all observations and existing data). See *Hyperphysics/Wikipedia*.

Problem Solving Questions

7 ${}^{40}_{19}K \rightarrow {}^{40}_{20}Ca + {}^{0}_{-1}b + \bar{n}$ (beta –) and ${}^{40}_{19}K \rightarrow {}^{40}_{18}Ar + {}^{0}_{+1}b + n$ (beta +)

8 Simply equate *A* and *Z* values both sides of the equation (${}^{7}_{4}Be + {}^{0}_{-1}b \rightarrow X + n$); answer is ${}^{7}_{3}$Li

9

- Equate *A* and *Z* numbers on LHS and RHS sides of the equation; gives *A* for Ba of **144**.
- 1 mole = 6×10^{23} ; multiply by 175 MeV and convert to J; this is about 17 TJ. A 1 GW power station gives out 3.6 TJ per hour.

10 Text does not mention *bremsstrahlung* but refers to *continuous* radiation – **see §40.6** – the incident electron transfers kinetic energy to the target, which is then re-emitted as x-ray photons This loss of kinetic energy of the electron is the "braking" caused by the target material. For the energy of the photon, use $\lambda = 35$ pm (from graph) – this is when the maximum kinetic energy (i.e. all) of the electron is transferred to the target and hence radiated x-ray photon. The two sharp peaks form the *characteristic x-ray spectrum*, caused when an electron in the target material is ejected by the incident electrons, and higher shell electrons quantum jump to fill the vacancy formed – the energy difference is

released as a characteristic photon. Difference peaks are caused by different source shells (e.g. n = 2 or n = 3 for K_α or K_β peaks respectively). For the K_α peak, transition between n = 2 to n = 1 is $\Delta E = E_2 - E_1 = -(13.6\text{eV})Z^2\left(\frac{1}{2}-\frac{1}{1}\right) = (10.2\text{eV})Z^2$

(For high Z elements, the energy equation $E_n = -\frac{(13.6\text{eV})Z^2}{n^2}$ is approximated to $-\frac{(13.6\text{eV})(Z-1)^2}{n^2}$ due to the effects of electron screening.)

11 Using a_B (the Bohr radius) = **5.29× 10^{-11} m**; r = **1.32 × 10^{-7} m**; E = **– 0.0218 eV**

Past Exam Questions

12 (a) Using $E_n = -\frac{(13.6\text{eV})Z^2}{n^2}$, N=1 is **–**13.6 eV, N=2 is –3.4 eV, N=3 is –1.5eV, N=4 is –0.85eV

12 (b) arrow points up from n= 1 to n = 4 **(c)** $\Delta E = E_4 - E_1 = (-13.6\text{eV})\left(\frac{1}{16}-\frac{1}{1}\right) = \mathbf{12.75\ eV}$

12 (d) ultraviolet ($\lambda = \frac{hc}{E} = \frac{4.14\times10^{-15}\times3\times10^{8}}{12.75} = \mathbf{974\ nm}$)

13 (a) $N_K = N_0 e^{-\lambda t}$ and $N_{Ar} = N_0 - N_0 e^{-\lambda t} \rightarrow N_0 = N_{Ar} + N_K$ … (I)

As $N_{Ar} = N_0 - N_0 e^{-\lambda t}$

$$\frac{N_0 - N_{Ar}}{N_0} = e^{-\lambda t}$$

$$\log_e\left(\frac{N_0 - N_{Ar}}{N_0}\right) = -\lambda t$$

Substitute in (I):

$$\log_e \frac{N_K + N_{Ar} - N_{Ar}}{N_k + N_{Ar}} = -\lambda t$$

$$-\log_e \frac{N_K}{N_k + N_{Ar}} = \lambda t$$

$$-\log_e \frac{N_K}{N_k + N_{Ar}} = \lambda t$$

$$\log_e \frac{N_K + N_{Ar}}{N_k} = \lambda t$$

$$\therefore \log_e\left(1 + \frac{N_{Ar}}{N_K}\right) = \lambda t$$

13 (b) First solve for decay constant $\lambda = \frac{\log 2}{\tau_{\frac{1}{2}}} = \frac{0.693}{1.25\times10^{9}\text{y}} = 5.54\times10^{-10}y^{-1}$

Thus, $t = \frac{1}{\lambda}\log_e\left(1 + \frac{10}{1}\right) = \mathbf{4.33\times10^{9}y}$

14(a) half-life of Uranium is of the same order or longer than the lifetime of the Earth (~4.5by); if it was much shorter it would all have vanished into daughter products.

14(b) beta (minus) decay (determined from the change in A and Z numbers)

14(c) Using $N = N_0 e^{-\lambda t} \rightarrow (1.00\mu\text{g})e^{-\left(\frac{0693}{432.2y}\right)30y} = 0.953\ \mu\text{g}$

14(d) ${}^{241}_{95}Am \rightarrow {}^{237}_{93}Np + {}^{4}_{2}\alpha$

15 (a)i. ${}^{99}_{42}Mo \rightarrow {}^{99m}_{43}Tc + {}^{0}_{-1}\beta + \bar{\nu}$ Beta-minus decay illustrated by A & Z numbers.

15 (a)ii. (Atomic mass of Mo) = 98.9077119u (Atomic Mass of TcM) = 98.9064065u

$$\rightarrow m_{Mo} = 98.9077119\text{u}\times931.49\frac{\text{MeV}}{\text{u}} = 92.131\ \text{GeV}$$

$$m_{Tc(m)} = 98.9064065\text{u}\times931.49\frac{\text{MeV}}{\text{u}} = 92.1303286\ \text{GeV}$$

$$\Delta E = (92.131545 - 92.1303286)\times10^{3}\ \text{MeV} = \mathbf{1.216\ MeV}$$

Note: Atomic masses *include* electron masses, so the mass of the beta particle is included within the value of the Tc(m) and does not need to be counted again.

15 (b) BE/c² = $\left[\left(44m_p + 55m_n + 44m_e\right) - m_{Ru}\right]$

$= [(1.65777\times10^{-25}kg) - (98.905393\times1.6605\times10^{-27}kg)]$

$\rightarrow$ BE $= 1.54\times10^{-27}\times c^2 = 1.3584\times10^{-10}$J $= 865.1$ MeV

$\therefore \frac{\text{BE}}{A} = \frac{865.1\text{ MeV}}{99} =$ **8.738 MeV per nucleon**

APPENDIX

When we wish to understand some aspect of the physical world, whether in this area of physics or any other, it is good practice to begin with what we call a "back of the envelope" calculation. This is any sort of very rough calculation which gives a useful initial impression of what might be going on, and some estimated order of magnitude.

To this end, let us consider a semi-classical model of hydrogen, as follows. Consider a classical particle of charge $-e$ moving at speed v in a circular orbit of radius r about a fixed charge $+e$. The force is $e^2/4\pi\epsilon_0 r^2$ so Newton's law "$F = ma$" gives

$$\frac{e^2}{4\pi\epsilon_0 r^2} = m\frac{v^2}{r} \qquad \Longrightarrow \qquad \frac{e^2}{4\pi\epsilon_0 r} = mv^2. \tag{3.1}$$

This shows that the potential energy is equal to minus twice the kinetic energy, and it is also a relationship between v and r. Next (this is the quantum part), we will use Heisenberg's Uncertainty Relation $\Delta x \Delta p \geq \hbar/2$. This is a precise statement about standard deviations, but in the present context we use it in a rough way. It is reasonable to guess that the radius r of the classical orbit gives the right order of magnitude for Δx, and similarly the value of v_x varies between $\pm v$ so mv gives an order of magnitude estimate for Δp. We argue that in general it is found that the wavefunction for the ground state in a single potential well gives spreads Δx and Δp such that $\Delta x \Delta p \simeq \hbar$. Thus we obtain

$$rmv \simeq \hbar. \tag{3.2}$$

Combining this with (3.1) we have

$$r \simeq \frac{4\pi\epsilon_0}{e^2}\frac{\hbar^2}{m} \simeq 0.5 \times 10^{-10}\,\text{m}$$

and

$$v = e^2/4\pi\epsilon_0\hbar \simeq c/137 \simeq 2000\text{ km/s}.$$

Hence the total energy is (P.E. + K.E.) = (−K.E.) = $-(1/2)mv^2 \simeq 13.6$ eV.